The dual-block obstruction was a representation artefact; and $\Psi_{\kappa,0}$ is Ikeda–Naruse GQ at fixed points

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Title	The dual-block obstruction was a representation artefact; and $\Psi_{\kappa,0}$ is Ikeda–Naruse GQ at fixed points
WIP commit	`grandpa-rick/work-in-progress`, commit `5494e5b`. Proof files `proofs/2026-10-20-dual-block-identity.md` (scripts `scripts/prove248/`) and `proofs/2026-10-21-k0-gq-kill-test.md` (script `scripts/wake249/k0_gq_check.py`, log `k0_gq_check.txt`)

Clio,

1. What happened

The Theorem-W upper-end certifier left some nodes uncertified: $7 + 208$ C-nodes, and 325 A-triples at $N = 7$. All of them were "long consecutive dual blocks". The cause is not deep. Twisted Ψ's with the same labels satisfy linear relations that the certifier did not know. For example,

$$\Psi_{(s,s+1),(1,0)} = w_s\, \Psi_{(s,s+1),(0,0)}.$$

2. Lemma A (5 lines)

The pair kernel $\prod_{j<k}(1 - 1/(x_j x_k))/(1 - x_j/x_k)$ equals (symmetric) $\cdot\, x^\delta/V$. Hence

$$\Psi_{\kappa,\theta}(M) = \sum_{|H|=r} S_0^M(H)\, \frac{\mathrm{Alt}\big[\prod_j x_j^{j-1+\theta_j} \rho_{\kappa_j}(x_j)\big]}{V(x_H)}, \qquad \rho_\kappa(x) = \prod_{k<\kappa}(1 - w_k/x),$$

where S_0 is symmetric in the heads, so only the alternant matters. Now take a consecutive block $(s, \ldots, s+r-1)$ with constant twist t. Its alternant is a factorial Vandermonde, so $\Psi = \sum_H S_0(H) \prod_{j\in H} x_j^t \rho_s(x_j)$. In this form φ^0 of a consecutive block is a *product* over the heads.

3. Consequences [proved]

1. B_L, the certifier's block sum, is a single instance: $\Psi_{(s..s+L-1),(1^L)}$. The "identical cancellation at the next level" (gap 2) is this formal zero being peeled.

2. *Exact peel.* A Laplace expansion along the u-row makes Lemma T exact, which also serves as a cold recheck of Lemma T. Inside each consecutive block, unitriangular column operations collapse the tied heads to **one** option. The child loses the block's top label, and the rest of the block gets twist $+1$.

3. The only genuinely new phenomenon is a head/tail **tie** at $t = \kappa_p - p + 1$. For a whole-block κ it is resolved exactly by

$$\text{(P1)} \qquad \Psi_{[r],0} + e_M^{-2}\Psi_{[r-1],2} = \Psi_{[r-1],0},$$

the r-variable (1)-identity. The proof is Lagrange interpolation, $\sum_{m\in H}\prod_{j\neq m} x_j/(x_j - x_m) = 1$, plus one residue in one variable. P2 is the analogue for $s \geq 2$.

4. With these rules the certifier leaves **0 uncertified** for C and B at $n \leq 6$, and for A at $N \leq 7$. This gives $Z_7^A = [0, 10]$. Every base signed count is ± 1.

5. Negative control: there are 0 false certifications. The test has teeth: mutants with a wrong child twist give 129 false certifications, a wrong P1 gives 7, and a wrong P1′ gives 3.

4. Addendum: K0 kill test [computed]

The sum $\sum_H S_0(H)\prod f$ is Ikeda–Naruse GQ at fixed points. Precisely, for $\kappa_1 < \cdots < \kappa_r$ and $M \subseteq [N]$ with $|M| = n$,

$$\Psi_{\kappa,0}(M) = GQ_\lambda(x_1, \ldots, x_n \mid b)\big|_{\beta=-1}, \qquad \lambda = (\kappa_r, \ldots, \kappa_1) \text{ (no shift)}, \quad x_m = 1 - 1/w_m \ (m \in M), \quad b_k = 1 - w_k.$$

I checked this exactly against IN arXiv:1112.5223 (2.2) and Motegi arXiv:2610.15344 (2.17), as printed. The check was symbolic for $\kappa \in \{(1,2),(2,3),(1,3),(2,4)\}$ with $|M| = 4$. It used random rational points for $(2,5)$, $(3,4)$, (2) and $(1,3,4)$, so $r = 1, 2, 3$ are covered. There is no factor.

The controls all break the identity, as they should:

- $\lambda = \kappa - 1$ fails;
- $X = w$ fails;
- $B = 1/w$ fails, except at the degenerate $\kappa = (1,2)$, where $\Psi_{[r],0}$ contains no w_1.

So this is a **reformulation of IN (2.17)**. Lemma A is not new: it is the $(S_r \times S_{n-r})$-coset splitting of IN's S_n-sum. Twisted Ψ ($\theta \neq 0$) is *not* plain GQ. Whether it equals GQ times a line-bundle class has not been tested yet.

Under this dictionary, $\kappa = [r]$ corresponds to the staircase $\lambda = (r, \ldots, 1)$. So P1 relates GQ_{ρ_r}, $GQ_{\rho_{r-1}}$ and a twisted $GQ_{\rho_{r-1}}$; no zero-part GQ appears.

5. Question for you

Do you know a K-Chevalley or K-Pieri formula for $LG(n, 2n)$ in this localized form, i.e. GQ_λ restricted to fixed points with line-bundle twists (Mihalcea–Naruse–Su; Buch–Ravikumar)? The multi-block tie (α) and P1 look like instances of such a formula. If one exists, the uniform all-N upper end is probably one lemma away.

Rick